DIBYA KIRTI MISHRA

Research Scholar
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CONTACT

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dibya kirti mishra

RESEARCH SKILLS

- **Image Proceesing**
  Feature identification and feature tracking using various image processing techniques in the ground and space-based data.
- **Machine Learning**
  Feature identification using Convolutional Neural Network (CNN) technique and other ML techniques (DBSCAN, GAN).

AREA OF RESEARCH

My research focuses on the long-term evolution of solar magnetic activity and its implications for solar dynamo theory and cycle prediction. I work extensively with century-long Ca II K and Hα datasets from the Kodaikanal Solar Observatory, complemented by observations from Mount Wilson Observatory, PSPT Rome, and space-based missions including the Solar Dynamics Observatory. I specialize in developing automated algorithms (Python/IDL) for feature detection and for constructing long-term time series from historical and modern solar data. My work includes:

- Long-Term analysis of chromospheric differential rotation using the KoSO Ca II K (1904–2007).
- Automated identification of polar network bright points, enabling the reconstruction of the historical polar magnetic field back to 1904.
- Machine-learning (CNN-based) detection of plages in century-long hand-drawn suncharts.
- Construction of long-term Carrington maps combining Ca II K data.
- Multi-wavelength studies of upper-atmospheric differential rotation using SDO/AIA.
- Development of ML-based pipelines for automated detection of sunspot groups and tilt angles from KoSO white-light data.
- Participation in a project to reconstruct pre-1976 magnetograms using Ca II K and Hα data through generative AI.

Overall, my expertise lies at the intersection of archival solar observations, automated feature-extraction techniques, and machine learning, aimed at improving our understanding of long-term solar variability and enhancing future solar-cycle predictions.

EDUCATION

Senior Project Associate (SPA) — **2026 - Now**
ARIES - Nainital, Uttarakhand (India)

Ph. D. - Solar Astrophysics — **2020 - 2025**
ARIES - Nainital, Uttarakhand (India)

Thesis Title - **Long Term Study of Sun using Space and Ground-based Data**
Thesis Supervisor - **Prof. Dipankar Banerjee**

M. Sc. - Physics — **2017 - 2019**
Central University of Kerala - Kasargod, Kerala

Passed with distinction.

B. Sc. - Physics — **2014 - 2016**
University of Delhi - Delhi

SKILLS

IDL	4+ yrs
Python	4+ yrs
Machine Learning	2+ yrs
LaTex	5+ yrs

ACHIEVEMENTS

CSIR-JRF

Physics
Qualified in 2019.

DST-SERB Travel grant

Astronomy
Obtained in 2024.

PROFILES

ORCID

OTHER SKILLS

Apart from my research work, I do sketching and outreach activities.

PUBLICATIONS

Differential Rotation of the Solar Chromosphere: A Century-long Perspective from Kodaikanal Solar Observatory Ca ii K Data — **2024**
Journal Name (ApJ) Vol 961, Issue 01, 2024, impact factor:5.4

Dibya Kirti Mishra, Srinjana Routh, Bibhuti Kumar Jha, Theodosios Chatzistergos, Judhajeet Basu, Subhamoy Chatterjee, Dipankar Banerjee, Ilaria Ermolli

Status: Accepted and Published

Exploring the dynamic rotational profile of the hotter solar atmosphere: A multiwavelength approach using SDO/AIA data — **2024**
Journal Name (ApJ) Vol 975, Issue 02, 2024, impact factor:5.4

Srinjana Routh, Bibhuti Kumar Jha, Dibya Kirti Mishra, Tom Van Doorsselaere, Vaibhav Pant, Subhamoy Chatterjee, Dipankar Banerjee

Status: Accepted and Published

Ca II K Polar Network Index of the Sun: a Proxy for Historical Polar Magnetic Field — **2025**
Journal Name (ApJ) Vol 982, Issue 02, 2025, impact factor:5.4

Dibya Kirti Mishra, Bibhuti Kumar Jha, Theodosios Chatzistergos, Ilaria Ermolli, Dipankar Banerjee, Lisa A Upton, M Saleem Khan

Status: Accepted and Published

Insights into Chromospheric Large-Scale Flows using Nobeyama 17 GHz Radio Observations I. The Differential Rotation Profile — **2025**
Journal Name (A&AL) Vol 982, 2025, impact factor:6.1

Srinjana Routh, Anshu Kumari, Vaibhav Pant, Jaydeep Kandekar, Dipankar Banerjee, Mohd. Saleem Khan, and Dibya Kirti Mishra

Status: Accepted and Published

Machine Learning Based Identification of Solar Disk and Plages in Kodaikanal Solar Observatory Historical Suncharts — **2025**
Journal Name (ApJS) Vol 283, Issue 01, 2026, impact factor:8.5

Dibya Kirti Mishra, Subhamoy Chatterjee, Bibhuti Kumar Jha, Hemapriya Raju, Aditya Priyadarshi, Manjunath Hegde, Srinjana Routh, Dipankar Banerjee, M Saleem Khan

Status: Accepted and Published

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PROCEEDINGS

Differential Rotation of the Solar Chromosphere using multidecadal Ca II K Spectroheliograms — **2024**
Journal Name (IAU Proceeding), 2024

Dibya Kirti Mishra, Srinjana Routh, Bibhuti Kumar Jha, Subhamoy Chatterjee, Dipankar Banerjee

Status: Accepted and Published

Estimation of Polar Magnetic Fields using Ca II K Polar Network as a Proxy **2024**
Journal Name (IAU Proceeding), 2025

Dibya Kirti Mishra, Bibhuti Kumar Jha, Theodosios Chatzistergos, Ilaria Ermolli, Dipankar Banerjee, M Saleem Khan

Status: Accepted

WORKSHOPS & CONFERENCES

Space radiation workshop **24-28 Jan 2022**
ARIES, Nainital, Uttarakhand, India

Online

40th Astronomical Society of India Meeting (ASI) **25-29 Mar 2022**
IIT Roorkee, Roorkee, Uttarakhand, India

Presented talk titled "Study of chromospheric Differential Rotation of the Sun using Ca II K Data".

National Supercomputing Mission (NSM) workshop on High Performance Computing (HPC) for Computational Fluid Dynamics (CFD) **17-20 May 2022**
IIT Bombay, Bombay, India

Participant

Aditya-L1 Science Support Cell Workshop **27 Jun-6 Jul 2022**
ARIES, Nainital, Uttarakhand, India

Volunteered as project guide.

Young Astronomers' Meet (YAM), 2022 **9-13 Nov 2022**
ARIES, Nainital, Uttarakhand, India

Presented poster titled "Differential Rotation in the Solar Chromosphere".

Aditya-L1 Science Support Cell (AL1SSC) second workshop **28-30 Nov 2022**
Manipal Academy of Higher Education (MAHE), Manipal, Karnataka, India

Participant

41st Astronomical Society of India meeting **01-05 Mar 2023**
IIT Indore, Indore, India

Presented talk titled "Variation in the Chromospheric Differential Rotation over the Century".

3rd BINA Workshop, Indo-Belgian Cooperation **22-24 Mar 2023**
Graphic Era Hill University, Bhimtal, Uttarakhand, India

Presented talk titled "Variation in the Chromospheric Differential Rotation over the Century".

SOLARNET Conference **8-12 May 2023**
Haus H, Telegrafenberg, Potsdam, Germany

Presented talk titled "Differential Rotation of the Solar Chromosphere using multidecadal Ca K spectroheliograms" (online).

Aditya-L1 Support Cell 4th Workshop **28 Jun-7 Jul 2023**
ARIES, Nainital, Uttarakhand, India

Volunteered as project guide.

International Astronomical Union Symposium (IAUS 365) **21-25 Aug 2023**
Yerevan, Armenia

Presented talk titled "Differential Rotation of the Solar Chromosphere using multidecadal Ca K spectroheliograms".

seventh Aditya-L1 Support Cell workshop **21-30 May 2024**
ARIES, Nainital, Uttarakhand, India

Presented talk and volunteered as project guide.

IAUG, 2024 **6-15 August 2024**
Cape Town, South Africa

Presented poster titled "Estimation of Polar Magnetic Fields using Ca II K Polar Network as a Proxy".

Joint Symposium of Space Climate 9 Symposium and ISEE Symposium **1-4 October 2024**
Nagoya University, Nagoya, Japan

Presented talk titled "Ca II K Polar Network as A Proxy for the Estimation of Historical Polar Magnetic Field of the Sun".

Solar cycle variability: From understanding to making prediction **14-18 October 2024**
ARIES, Nainital, Uttarakhand, India

Presented talk titled "Ca II K Polar Network as a Proxy to Estimate the Polar Field in the Sun".

Sun, Space Weather, and Solar-Stellar Connection **20-24 January 2024**
IIA, Bengaluru, India

Presented poster titled "Automated Detection of Plages in Hand-drawn Suncharts from the Kodaikanal Solar Observatory Using Machine Learning Technique".

OTHER RESPONSBILITIES

- Organisation and structuring of data observed from H α telescope, ARIES for the maintenance of the archival site.
- Active participation in Aditya-L1 workshops as a project mentor.

PARTIAL GUIDANCE

- **Amrutha P**, currently a MSc student at the Amrita Vishwa Vidyapeetham (Amritapuri Campus), Kollam, Kerala, India (**Summer Research Fellowship Program**).
- **Debargha Biswas**, currently a PhD student at IIST, Trivandrum, Kerala, India.